

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED  
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

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| <b>Sec No-<br/>10</b> | <b>Team No- 20</b> | <b>Project Title- Enterprise banking micro services</b> |
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**1. Abstract**

Enterprise Banking Microservices is a distributed backend system designed to provide scalable, secure, and reliable banking services using a microservice architecture. The system divides major banking operations such as customer management, account management, transactions, payments, and authentication into independent services. Each microservice can operate and scale independently while communicating through well-defined APIs. A distributed database approach is used to maintain data consistency, reliability, and efficient access to banking information. The project demonstrates important database engineering concepts such as transaction management, concurrency control, data integrity, service-level database design, and secure data access. The architecture also supports fault isolation and scalability, making it suitable for handling large numbers of users and banking transactions. Overall, the project provides practical experience in designing and implementing a distributed banking backend using modern microservices and database technologies.

**2. Team Details**

| <b>Roll Number(s)</b> | <b>Name of the Student</b> | <b>Signature</b> |
|-----------------------|----------------------------|------------------|
| 2520090160            | Snigdha                    |                  |
| 2520090161            | Koumudi                    |                  |
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### 3. Frameworks / Platforms / Software Used

- **Backend Framework:** Spring Boot / Node.js with Express
- **Programming Language:** Java / JavaScript
- **Database:** PostgreSQL / MySQL
- **Microservices Architecture:** Spring Cloud
- **API Communication:** REST APIs
- **API Testing:** Postman
- **Service Discovery:** Eureka
- **API Gateway:** Spring Cloud Gateway
- **Containerization:** Docker
- **Version Control:** Git and GitHub
- **Development Environment:** IntelliJ IDEA / VS Code
- **Database Management:** MySQL Workbench / pgAdmin
- **Build Tool:** Maven
- **Authentication & Security:** Spring Security, JWT
- **Documentation:** Swagger / OpenAPI

Guide Name and Signature (remarks if any) \_\_\_\_\_